STARWEAVER®

Hands-On-Learning: Writing an Investigative Report

1. Lab Overview

This lab brings together the skills you have developed in Modules 3 and 4. You will take on the role of the lead investigator who has just completed the analysis in the Module 3 lab ("Log Analysis and Attack Correlation"). Your task is to now formally document your findings in a professional, defensible forensic report suitable for a non-technical audience (e.g., the management of "Innovate Solutions Inc.").

2. Prerequisites

Before starting this lab, you should have reviewed all nine video scripts for Module 4, which cover forensic report writing, creating demonstrative evidence, preparing for testimony, and the professional responsibilities of an investigator.

3. Learning Objectives

- Apply a structured format to write a formal investigative report.
- Translate technical findings into a clear, non-technical narrative.
- Draft a concise executive summary that communicates the most critical findings.
- Properly cite evidence to support factual conclusions.

4. Scenario

To refresh your memory, here is the scenario from the Module 3 lab:

A small business, "Innovate Solutions Inc.," experienced a security incident. The payroll manager, Bob, received a phishing email. The next day, an unauthorized login to the company's payroll portal was detected. Your analysis of the email header, firewall logs, and a cryptocurrency transaction note produced the following key findings:

- The Phishing Email: The email, purporting to be from "SecureNet IT Support," was a fake. The From address was spoofed. The email's true origin was an IP address: 198.51.100.24.

- The Initial Compromise: The firewall logs show that shortly after the email was received, Bob's workstation (10.0.1.55) made a connection to the attacker's IP address (198.51.100.24). This strongly suggests the phishing link was clicked and a malicious payload may have been delivered or credentials were stolen.
- The Unauthorized Login: The next day, the firewall logs show a successful connection to the company's public payroll portal (209.10.20.30) from the attacker's IP address (198.51.100.24).

- The Financial Action: Two days after the breach, a cryptocurrency transaction was noted, indicating a potential extortion payment or money laundering activity connected to the attacker.

Your job is to take these facts and present them in a formal report.

5. Tools and Resources

- A text editor (like Notepad, VS Code, or TextEdit) to write your report in Markdown format.
- Your notes and the video script from Module 4, Lesson 1 on writing technical reports.

6. Step-by-Step Walkthrough

1. Create a New Document: Create a new text file and name it investigative_report.md.
2. Structure Your Report: Using the structure from the lesson, create the following sections in your document using Markdown headers:

```
# Investigative Report: Innovate Solutions Inc. Payroll Incident
## 1. Executive Summary
## 2. Case Information
## 3. Objectives of the Investigation
```

4. Evidence Examined

5. Timeline of Events

6. Findings

7. Conclusion

3 Flesh out each section:

Executive Summary: Write a brief, one-paragraph summary. State the conclusion upfront: the payroll portal was breached by an attacker who initiated the attack with a phishing email. Mention the attacker's IP.

Case Information: Fill in some basic details (e.g., Case #: IS-2025-0516, Investigator: Your Name, Date: Today's Date).

- Objectives: State the goal, e.g., "To analyze the provided evidence to determine the source of the unauthorized login to the payroll portal."
- Evidence Examined: List the evidence you were given (e.g., email_header.txt, firewall_log.csv).
- Timeline of Events: This is the core narrative. Create a chronological list of what happened, with timestamps. Translate the technical events into plain English.
  - Example: "May 15, 2025, 2:30 PM: An employee received a phishing email from an attacker at IP address 198.51.100.24."
- Findings: List your key conclusions as factual statements, referencing the evidence.

Example: "Finding 1: The email received by the employee was a phishing attack, confirmed by analyzing the email header which showed the true origin IP."

Example: "Finding 2: The attacker's IP address (198.51.100.24) was responsible for both the initial phishing email and the subsequent unauthorized login to the payroll portal, as shown in the firewall logs."

Conclusion: Restate your main conclusion. It is your professional opinion, based on the evidence, that the attacker used credentials obtained from the phishing attack to access the payroll portal.

7. Deliverable: The Final Report

Your completed investigative_report.md file is the deliverable for this lab. It should be a complete, professional document that clearly explains the incident to a non-technical reader.

8. Assessment Rubric

Criteria	Not Met (0 pts)	Partially Met (5 pts)	Fully Met (10 pts)
Report Structure	The report is missing key sections or is poorly organized.	The report has most sections but they are incomplete or out of order.	The report is well-structured with all required sections present and logically ordered.
Executive Summary	The summary is unclear, inaccurate, or contains jargon.	The summary explains the incident but is not concise or clear.	The summary is clear, concise, accurate, and perfectly suited for a non-technical executive.

8. Assessment Rubric

Criteria	Not Met (0 pts)	Partially Met (5 pts)	Fully Met (10 pts)
Clarity of Narrative	The timeline and findings are confusing and filled with jargon.	The report explains what happened, but the language is still somewhat technical.	The report uses plain, accessible language to create a clear and easy-to-follow narrative of the attack.

8. Assessment Rubric

Criteria	Not Met (0 pts)	Partially Met (5 pts)	Fully Met (10 pts)
Evidence-based	Conclusions are stated without reference to the evidence.	Some findings are linked to evidence, but not all.	Every major finding and timeline event is explicitly supported by a reference to the evidence (e.g., email header, firewall log).

9. Reflection Questions

What was the most challenging part of translating the raw log data into a formal report?

If you were presenting this report to the company's board, what is the single most important sentence you would want them to remember?

How does writing a detailed, evidence-based report like this prepare you for a potential cross-examination in court? Why is it important to separate the "Timeline" (what happened) from the "Findings" (what it means) in that context?

10. Additional Resources

SANS Institute - Writing Forensic Reports

NIST Guide to Integrating Forensic Techniques into Incident Response (SP 800-86)

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

Share the Project

Document: Save the report in PDF format.

Upload: Share in the "Peer Review" area of the course shell with a brief description.

Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.